Exact Label-Allocation Audit of Frobenius Splitting Scores

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Abstract

Using previously verified scores, we exactly count all 20037600 fixed-field label assignments for a reported comparison of 42 additional prime-order candidates with eight publications. Of 102 within-field pairs, 55 have the additional candidate higher, 7 tie and 40 are lower. The inclusive upper allocation tail is $5869345/20037600 \simeq 0.292916567$, consistent with the source's Monte Carlo estimate near 0.29. Independent subset-enumeration and rank-sum implementations agree on all five output files. No new curve is sampled. This is a retrospective arithmetic audit, not a historical selector, real-curve null calibration or ECDLP attack.

1 Fixed comparison and source scope

For a curve over $\mathbb{F}_p$ of full prime order n, let $\Delta = (p+1-n)^2 - 4p$ and count odd primes $\ell \le 691$ with $(\Delta/\ell) = +1$. Zero characters are separate. The earlier independently verified census retains 69 seed positions and 82 branches [1]. Splitting alone gives no strict separator over that broader census at any cutoff from 100 through 5000; its original 21-choice subset still reproduces the narrow splitting pattern.

An external report compares the 42 additional root/width-category candidates in the 192-, 224-, 239- and 256-bit fields with the eight same-field publications [2]. We retain that definition and cutoff. The eleven named omissions, seven P-384 positions and singleton P-521 remain in our input but are outside this particular 50-position comparison. Its data and labels were already visible before this audit, so it is not a fresh holdout for us.

All 60 seed-identified external observations match our saved scores. Eight source publication rows lack seed identifiers; their scores are checked as within-field multisets, without claiming individual curve-identity replay. Source model percentiles are not used. Source deviations—missing checkpoints, unretained 48-bit validation rows, and a noncanonical protocol freeze/export— remain disclosed; this later audit repairs none of that history [3].

2 Exact allocation arithmetic and result

Let U award one per pair when the additional candidate's score is higher, one half at a tie, and zero otherwise. We use the integer $2U$ throughout. The conditional allocation inventory is

$$\binom{24}{3}\binom{9}{1}\binom{12}{3}\binom{5}{1} = 20037600.$$

The producer enumerates each field's publication subsets and convolves their integer histograms. An independent verifier uses doubled midranks r_i and a chosen-rank-sum dynamic program. With K publications and H additional positions in a field, $2U = 2KH + K(K+1) - \sum_{i\in\text{published}} r_i$. Both methods retain tied values as distinct labelled positions.

Field bits	Published	Additional	Pairs	Higher	Tied	Lower
192	3	21	63	29	6	28
224	1	8	8	7	1	0
239	3	9	27	17	0	10
256	1	4	4	2	0	2
Total	8	42	102	55	7	40

"Higher" means additional > published. The observed fraction is $(55+7/2)/102 = 117/204 \simeq 0.573529412$. Exactly 5869345 assignments have $2U \ge 117$, giving the tail above. It does not meet the source's nominal 0.05 replication threshold within this allocation model. A p-value interpretation requires within-field label exchangeability; the exact finite fraction itself assumes no real-curve sampling model. It neither proves an absence of selection nor supplies a probability of historical manipulation.

3 A public generator relationship is not a solver

Suppose $G, Q \neq O$ lie in the same cyclic group of prime order n, with a known relation $G = [s]Q$, $s \not\equiv 0 \pmod{n}$. For a target $P = [k]G$,

$$[s^{-1} \bmod n]P = [k]Q.$$

Conversely a target $R = [k]Q$ maps to $[s]R = [k]G$. These public invertible maps convert the DLP between bases; they do not supply the destination solver. If a recipe derives s from b, its conversion from $\mathbb{F}_p$ to a scalar modulo n must be specified. If $s = 0$, it yields the identity and fails the nonidentity-generator premise. This equivalence is not a general security proof or a measured attack.

For $a = -3$, coordinates satisfy $b = y^2 - x^3 + 3x \pmod{p}$: $x = 0$ gives $y^2 = b$, and $x = 1$ gives $y^2 = b - 2$. These field identities do not by themselves give scalar logarithms. The supplied canonical points were assigned by a public first-x rule; they do not reveal historical unpublished generators. RFC 5639 Appendix A.2, steps 12–13, gives a later standard example of a smallest-x point multiplied by a derived scalar [4]. Its subsequent small-coordinate side-channel discussion concerns implementation behaviour. It is not evidence that the 1997 NIST program used that construction.

4 Verification and remaining evidence gap

Source and input bindings were committed before allocation arithmetic at `fc7bac88dc217d01c59ff750d973d2796ebba1b4`. Separate C++17 programs use subset enumeration and rank dynamic programming; only the compiler and TSV I/O are shared. All five files agree byte for byte. All ten phases pass as specified: seven native zero exits and three deliberate corrupt-output refusals with exit one. Full field/joint distributions, source checks, commands, binaries and native statuses are retained. No new character, point count, factorization, random draw, DLP or performance measurement is involved.

The exact existing-results audit uses the repository's local development exception, explicitly recorded before execution. No checkpoint, S3 receipt or gate on the separate sampling experiment is claimed passed. The requested 10000 accepted real prime-order curves per field and publication percentiles remain unrun. The supported launch route still lacks a qualified PARI worker. No original candidate visitation/acceptance logs are available. The combined root/width/splitting fit, both weakness hypotheses and the historical cause remain unresolved. No finite trait catalog exhausts all mathematical functions.

References

[1] Prior verified census and scope report. Crypto revision `c6d8897d7582fed39e8eb7fdd4e312a70d692532`, `research/nist_skip_criterion_20261025/broader_census_reanalysis_20261030`.

[2] Prior lab report. Small-prime splitting: null, cutoff sweep and held-out test. Crypto-autoresearcher revision `3ff8c32c15f91b355d18e207d6044a1d2363499f`, `research/nist_seed_split_null_20261029/report.md`.

[3] Same source revision and study, `DEVIATIONS.md`, deviations D1–D3. Original measurement admission is not retroactively repaired.

[4] M. Lochter and J. Merkle. RFC 5639, Appendix A.2, March 2010. `https://www.rfc-editor.org/rfc/rfc5639.txt`.